Optimal Reconfiguration of Systems in Aeronautics

Lecture 06: "Schur Vector Method for the Algebraic Riccati Equation"

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Outline

1. Direct Methods for Fault Accommodation
2. B747 Application Example

Cuprins

1. Direct Methods for Fault Accommodation
   - Preliminaries on Numerical Algorithms
   - Schur Vector Method
   - Generalized Schur Vector Method
   - Eigenvector Method
2. B747 Application Example

Objective I

This lecture provides some results on numerical algorithms.

It is mainly used when the Schur Vector Method to solving the post-fault Continuous Algebraic Riccati Equation is presented.

It gives the definition of elementary reflectors (Householder transformations), provides details on the application of the `QR`-algorithm, and makes some considerations about the recently developed structure-preserving numerical transforms.

Elementary reflectors I

Lemma (Real Householder Transformation)

Given the column vector $x_f \in \mathbb{R}^n$, *define*

$$\tilde{u}_f = x_f \pm \|x_f\|_2 e_1 \quad \textit{and} \tag{1}$$

$$U_f = I_n - 2\frac{\tilde{u}_f \tilde{u}_f^\top}{\tilde{u}_f^\top \tilde{u}_f}\,, \tag{2}$$

where $\|x_f\|_2$ *stands for the* 2*-norm of* x_f *and* e_1 *indicates the first identity vector from the standard basis of* $\mathbb{R}^n$.

Then it holds that

$$U_f x_f = \mp \|x_f\|_2 e_1\,. \tag{3}$$

□

Elementary reflectors II

Usually, (1) gets concrete as follows

$$\tilde{u}_f = x_f + \text{sign}\,(x_{f_1})\|x_f\|_2 e_1\,, \tag{4}$$

and (3) results in zeroing all but the first component of x_f.

The choice of the sign-function in (4) is dictated by the need to avoid a small Euclidean-norm of $\tilde{u}_f$ in (1), as it appears in the denominator of (2).

Such matrix U_f is called an elementary reflector or Householder matrix; it is orthogonal and symmetric ($U_f^\top U_f = U_f U_f^\top = I_n$, $U_f = U_f^\top$).

QR-algorithm I

It assumes the QR-factorization of the Hamiltonian-matrix H_f at iteration $k-1$, i.e.,

$$H_{f_{k-1}} = Q_{f_k} R_{f_k} \tag{5}$$

and iterate conversely at the current step as

$$R_{f_k} Q_{f_k} = H_{f_k}\,; \tag{6}$$

matrices Q_{f_k} are products of Householder-transforms ($Q_{f_k} = \prod_{j=1}^{k} U_{f_k}^{\top}$), computed as in (2), with the difference that index 1 is replaced in (4) by index j, such that the effect is the vector-segment from $j+1$ to n gets zero in the leading H_{f_j} (the vector e_j in (4) is the j-th unit-vector from the standard basis of $\mathbb{R}^n$, that is a vector with a 1 on its j-th position and 0's elsewhere).

QR-algorithm II

However, do not forget that a preliminary step to (5)-(6) is the reduction of the initial matrix H_f to the upper-Hessenberg form ($H_{f_{i,j}} = 0$, for $i > j + 1$); the upper-Hessenberg form remains invariant under the QR-iteration and, after a finite number of steps, the series of matrices H_{f_k} converges to the Real Schur Form (RSF) of the initial Hamiltonian-matrix H_f.

This presentation of the QR-algorithm is simplistic, because other numerical processes are usually involved, like shifting and deflation, as well.

Structure-preserving numerical transforms I

One way to accumulate the Householder matrices is given by the WY-representation, that is computing the orthogonal transforms U_f at the current step k as

$$U_{f_k} = \prod_{j=1}^{k} U_{f_j} = I_n + W_f Y_f^\top \ , \tag{7}$$

where W_f and Y_f are $n \times k$ real matrices and Y_f is lower trapezoidal ($Y_{f_{i,j}} = 0$, if $i < j$).

Its extension, called the compact WY-representation, assumes computing U_{f_k} as

$$U_{f_k} = \prod_{j=1}^{k} U_{f_j} = I_n + Y_f T_f Y_f^\top \ , \tag{8}$$

where Y_f and T_f are $n \times k$ and $k \times k$, respectively, real trapezoidal and upper-triangular matrices.

Both approaches concern the aggregation of the elementary reflectors and the arrangement of operations such that mostly level 3 BLAS (Basic Linear Algebra Subprograms) matrix-matrix multiplications to be involved.

Also, following the lines of the UT-transform proposed recently could represent another efficient approach.

Structure-preserving numerical transforms II

However, performing the QR-iterations on H_f as it is shown above ignores somehow its Hamiltonian property and provides to be inefficient too.

This may be avoided by using structure-preserving block-matrix algorithms based on orthogonal symplectic similarity transformations.

Recall that a matrix H_f in $\mathbb{R}^{2n\times 2n}$ is said to be J-orthogonal or symplectic if

$$H_f^\top J H_f = J\,, \tag{9}$$

where J is the classical ubiquitous matrix.

Real Schur form I

Remember that, given the real general square-matrix $H_f \in \mathbb{R}^{n \times n}$, it is said to be similar with matrix $S_f \in \mathbb{R}^{n \times n}$, if they share the same spectrum, $\Lambda(H_f) = \Lambda(S_f)$, where the spectrum is the set of matrix eigenvalues, i.e., the roots of the characteristic polynomial, precisely of the equation $\det(\lambda I_n - H_f) = 0$, where $\det(.)$ denotes the matrix determinant.

Hence, the Real Schur Decomposition as an important result on similarity transformations is introduced.

Real Schur form II

Theorem (Real Schur Decomposition)

Given the matrix $H_f \in \mathbb{R}^{n \times n}$, there exists an orthogonal matrix U_f so that

$$S_f = U_f^\top H_f U_f$$

is in (quasi-)triangular form

$$S_f = \begin{bmatrix} S_{f_{11}} & \star & \dots & \star \\ O & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \star \\ O & \dots & O & S_{f_{pp}} \end{bmatrix};$$

on the main diagonal of S_f, the Real Schur Form (RSF) of H_f, block-matrices $S_{f_{ii}}$ ($i = 1, 2, \dots, p$, with $\dim(S_{f_{11}}) + \dots + \dim(S_{f_{pp}}) = n$) of order 1×1 or 2×2 appear, as they correspond to the real respectively the pairs of complex-conjugate eigenvalues of H_f, where $\dim(.)$ gives the order of the matrix-argument.

In addition, U_f can be chosen so that S_f is in RSF and its diagonal blocks appear in any desired order. (Symbols $\star$ indicate nonzero elements.) $\square$

Schur vector method I

Also, reconsider the LTI system (10), respectively

$$M_f: \quad \dot{x}(t) = A_f x(t) + B_f u_f(t)\,, \text{ with } x_{t_f} = x(t_f) \triangleq x_f\,, \tag{10}$$

along with the minimization of the quadratic cost functional (11), as

$$J[x_f, u(t)] = \int_{t_f}^{+\infty} \left[x^\top(t) Q_n x(t) + u^\top(t) R_n u(t)\right] dt\,, \tag{11}$$

and analyze the optimality of (12), under the form

$$u_f(t) = -K_f x(t)\,. \tag{12}$$

Schur vector method II

By analogy with the Pontryagin Maximum Principle (PMP), write the control-dependent Hamiltonian function for the post-fault system

$$h_{u_f}[\mu(t), x(t)] = \mu^\top(t)\left[A_f x(t) + B_f u_f(t)\right] + \left[x^\top(t) Q_n x(t) + u_f^\top(t) R_n u_f(t)\right] \tag{13}$$

$$= \left[\mu^\top(t) A_f + x^\top(t) Q_n\right] x(t) + \left[\mu^\top(t) B_f + u_f^\top(t) R_n\right] u_f(t)\,, \tag{14}$$

where $\mu^\top(t) \in \mathbb{R}^n$ represents the co-state (adjoint) vector, which is defined as follows

$$\dot{\mu}(t) = -\left\{\frac{\partial h_{u_f}[\mu(t), x(t)]}{\partial x(t)}\right\}^\top ;$$

that yields

$$\dot{\mu}(t) = -Q_n x(t) - A_f^\top \mu(t)\,; \tag{15}$$

(remember that $Q_n = Q_n^\top$).

Schur vector method III

In (13), the second term depending on the control variable $u_f(t)$ has a unique minimum at the point where

$$\frac{\partial h_{u_f}[\mu(t), x(t)]}{\partial u_f(t)} = 0\,,$$

and thus

$$u_f(t) = -R_n^{-1} B_f^\top \mu(t)\,; \tag{16}$$

(remember that $R_n = R_n^\top$ too).

Schur vector method IV

Now, since the minimized Hamiltonian

$$h^*_{u_f}[\mu(t), x(t)] = \min_{u_f \in \mathbb{R}^m} h_{u_f}[\mu(t), x(t)]$$

in (13) is smooth, extremals are solutions of the corresponding Hamiltonian system based respectively on (15) and (16), precisely

$$\dot{x}(t) = A_f x(t) - B_f R_n^{-1} B_f^\top \mu(t)\,, \tag{17a}$$

$$\dot{\mu}(t) = -Q_n x(t) - A_f^\top \mu(t)\,, \tag{17b}$$

with conditions at the boundary given by $x(t_f) = x_f$ and $\mu(+\infty) = 0$.

Then, construct its corresponding Hamiltonian-matrix

$$H_f = \begin{bmatrix} A_f & -G_f \\ -Q_n & -A_f^\top \end{bmatrix}\,, \tag{18}$$

where

$$G_f = B_f R_n^{-1} B_f^\top\,.$$

Schur vector method V

Furthermore, recall that applying the QR-algorithm on the Hamiltonian-matrix (18) drives it to the upper (quasi-)triangular Real Schur Form (RSF).

Then, use the previous theorem and write the RSF of H_f as

$$S_f = U_f^\top H_f U_f,$$

where $H_f \in \mathbb{R}^{2n\times 2n}$ this time, and remember that matrices H_f and S_f above are similar; assume also the ordering of the block-matrices on the diagonal of S_f such that in the upper-left corner the stable eigenvalues reside (if H_f has no eigenvalues on the imaginary axis, then a stabilizing solution to CARE$_f$ still exists, the condition of H_f dichotomic being necessary but however not sufficient).

Schur vector method VI

Also, after the QR-algorithm reached convergence, partition S_f and the matrix of accumulated transformations U_f conformably with H_f as

$$S_f = \begin{bmatrix} S_{f_{11}} & S_{f_{12}} \\ O & S_{f_{22}} \end{bmatrix} \text{ and } U_f = \begin{bmatrix} U_{f_{11}} & U_{f_{12}} \\ U_{f_{21}} & U_{f_{22}} \end{bmatrix},$$

and note that the first n columns of U_f, precisely $\begin{bmatrix} U_{f_{11}}^\top & U_{f_{21}}^\top \end{bmatrix}^\top$, span the stable invariant-subspace of the Hamiltonian-matrix (18).

In this case, the stabilizing solution to CARE$_f$ computes as

$$X_f = U_{f_{21}} U_{f_{11}}^{-1} \tag{19}$$

and

$$\Lambda(S_{f_{11}}) = \Lambda(A_f - G_f X_f) \equiv \Lambda(A_f - B_f K_f) \subset \mathbb{C}^- ,$$

where $\Lambda(.)$ stands for the spectrum of the matrix-argument.

Schur vector method VII

This is to be seen by considering the invariant subspace equation

$$H_f \begin{bmatrix} U_{f_{11}} \\ U_{f_{21}} \end{bmatrix} = \begin{bmatrix} U_{f_{11}} \\ U_{f_{21}} \end{bmatrix} S_{f_{11}}.$$

Recall that, given H_f a matrix in $\mathbb{R}^{n\times n}$, a subspace $\mathcal{S}_f$ of $\mathbb{R}^n$ ($\mathcal{S}_f \subset \mathbb{R}^n$) is said to be H_f-invariant if $H_f\mathcal{S}_f \subseteq \mathcal{S}_f$, where $H_f\mathcal{S}_f = \{x_f \in \mathbb{R}^n \mid x_f = H_f y_f, \text{ for some } y_f \in \mathcal{S}_f\}$. Remember also that a set $\mathcal{S}_f$, subspace of $\mathbb{R}^n$, is said to span the n-th order real vector-space if $\text{span}\,\mathcal{S}_f = \mathbb{R}^n$, where the span of $\mathcal{S}_f$ is the subspace of $\mathbb{R}^n$ of all linear combinations of $s_{f_1}, s_{f_2}, \ldots, s_{f_k} \in \mathcal{S}_f$.

Schur vector method VIII

This far, assuming that the average QR-iterations per eigenvalue is 2 and including the RSF ordering of the Hamiltonian-matrix, the work-count for the Schur Vector Method is estimated at nearly $200\,n^3$ flops.

Remember that a floating-point operation, flop in short, stands for the arithmetic effort in computing $a \leftarrow a + b \times c$, that is calculating and updating the result of an addition and a multiplication, where a, b, c are floating-point numbers.

Generalized Schur vector method I

Since working with the Hamiltonian-matrix (18) involves handling R_n^{-1}, numerical problems may appear if matrix R_n is ill-conditioned, while this can be avoided by analogy with the Schur vector method.

Look again at the Hamiltonian system (17) and extend it as

$$\dot{x}(t) = A_f x(t) + B_f u_f(t)\,, \tag{20a}$$

$$\dot{\mu}(t) = -Q_n x(t) - A_f^\top \mu(t)\,, \tag{20b}$$

$$0 = R_n u_f(t) + B_f^\top \mu(t)\,, \tag{20c}$$

thus transforming the original two-point boundary-value problem into the following

$$\begin{bmatrix} I_n & O_n & O_n \\ O_n & I_n & O_n \\ O_n & O_n & O_n \end{bmatrix} \begin{bmatrix} \dot{x}(t) \\ \dot{\mu}(t) \\ \dot{u}_f(t) \end{bmatrix} = \begin{bmatrix} A_f & O_n & B_f \\ -Q_n & -A_f^\top & O_n \\ O_n & B_f^\top & R_n \end{bmatrix} \begin{bmatrix} x(t) \\ \mu(t) \\ u_f(t) \end{bmatrix}, \tag{21}$$

with conditions at the boundary given by $x(t_f) = x_f$ and $\mu(+\infty) = 0$, and $u_f(t)$ minimum along the optimal trajectory.

Generalized Schur vector method II

Therefore, obtaining the solution to CARE$_f$ by computing the stable deflating-subspace of an extended $(2n+m)\times(2n+m)$ real Hamiltonian regular pencil

$$\bar{H}_f = \lambda M_f - N_f, \text{ with } \det(\lambda M_f - N_f) \neq 0\,, \tag{22}$$

for matrices M_f and N_f from (21) as

$$M_f = \begin{bmatrix} I_n & O_n & O_n \\ O_n & I_n & O_n \\ O_n & O_n & O_n \end{bmatrix} \text{ and } N_f = \begin{bmatrix} A_f & O_n & B_f \\ -Q_n & -A_f^\top & O_n \\ O_n & B_f^\top & R_n \end{bmatrix},$$

is recommended.

A QZ-transform applied on the matrix-pencil $\bar{H}_f$ in (22) provides its generalized Schur factorization, and thus its stable deflating-subspace.

Generalized Schur vector method III

Once again, partitioning the orthogonal factor V_f conformably with $\bar{H}_f$, its first n columns, precisely $\begin{bmatrix} V_{f_{11}}^\top & V_{f_{21}}^\top & V_{f_{31}}^\top \end{bmatrix}^\top$, span the deflating-subspace for stable eigenvalues.

In this situation, the stabilizing solution to CARE$_f$ calculates similarly with (19) as follows

$$X_f = V_{f_{21}} V_{f_{11}}^{-1}, \tag{23}$$

and the post-fault control matrix K_f computes like in below

$$K_f = V_{f_{31}} V_{f_{11}}^{-1}; \tag{24}$$

however, it appears that the relationship of the post-fault control matrix K_f with the solution X_f to the post-fault CARE given in (23) is somehow lost as it may be computed directly as in (24), as well.

Note also that the Schur Vector Method as well as the Generalized Schur Vector Method might be numerically unstable due to the possible ill-conditioning of matrix $V_{f_{11}}$.

Eigenvector method I

Alternatively, a similar expression for X_f may be established by using the general Eigenvector Method.

In this case, the solution X_f to the post-fault CARE writes as

$$X_f = Y_{f_2} Y_{f_1}^{-1} ,$$

where $Y_{f_{1,2}}$ are real $n \times n$ matrices formed by the upper- and lower- halves of eigenvectors or generalized-eigenvectors in a Jordan chain of H_f chosen such that Y_{f_1} is invertible.

Though the flop-count for this approach is estimated at a lower $160\, n^3$, it appears that the Eigenvector Method may become unstable for multiple eigenvalues and the Schur Vector Method is better numerically conditioned.

Still, to solving the post-fault CARE (25), respectively

$$CARE_f : \quad A_f^\top X_f + X_f A_f - X_f B_f R_n^{-1} B_f^\top X_f + Q_n = 0 , \tag{25}$$

it is worth investigating next the Newton-Kleinman and Matrix-Sign-Function techniques as iterative alternatives to the direct methods.

Cuprins

1 Direct Methods for Fault Accommodation

2 B747 Application Example
- Direct Fault Accommodation - Schur Vector Method

Direct fault accommodation - Schur vector method I

Recall first to form the Hamiltonian-matrix

$$H_f = \begin{bmatrix} -0.2814 & 696.6000 & -3.2190 & -0.2077 \\ -0.0009 & -0.3856 & -0.2077 & -0.0134 \\ -0.0100 & 0.0 & 0.2814 & 0.0009 \\ 0.0 & -0.0100 & -696.6000 & 0.3856 \end{bmatrix},$$

with

$$G_f = \begin{bmatrix} 3.2190 & 0.2077 \\ 0.2077 & 0.0134 \end{bmatrix}.$$

The Real Schur Form (RSF) of H_f is

$$S_f = \begin{bmatrix} -1.9583 & -3.0017 & 627.3972 & -304.2751 \\ 1.4431 & -1.9583 & 301.0627 & 627.3972 \\ 0.0 & 0.0 & 1.9583 & 3.0017 \\ 0.0 & 0.0 & -1.4431 & 1.9583 \end{bmatrix},$$

for the accumulated orthogonal similarity transformation ($U_f^\top U_f = I_4$), namely the matrix

$$U_f = \begin{bmatrix} -0.9751 & -0.2217 & -0.0023 & 0.0028 \\ 0.0018 & 0.0050 & -0.9751 & 0.2217 \\ -0.0028 & 0.0023 & -0.2217 & -0.9751 \\ -0.2217 & 0.9751 & 0.0050 & 0.0018 \end{bmatrix}.$$

Direct fault accommodation - Schur vector method II

Extract from U_f the following block-matrices

$$U_{f_{21}} = \begin{bmatrix} -0.0028 & 0.0023 \\ -0.2217 & 0.9751 \end{bmatrix} \text{ and } U_{f_{11}} = \begin{bmatrix} -0.9751 & -0.2217 \\ 0.0018 & 0.0050 \end{bmatrix},$$

and calculate the inverse of the latter as

$$U_{f_{11}}^{-1} = \begin{bmatrix} -1.1167 & -49.1565 \\ 0.4011 & 216.1851 \end{bmatrix};$$

(however, the inverse $U_{f_{11}}^{-1}$ should not be used but the solution of n linear systems with the same coefficient matrix, i.e., $X_f = U_{f_{21}} / U_{f_{11}}$).

Direct fault accommodation - Schur vector method III

Furthermore, compute X_f upon (19) as

$$X_f = \begin{bmatrix} 0.0040 & 0.6387 \\ 0.6387 & 221.7001 \end{bmatrix},$$

and note that the spectrum of the stable closed-loop matrix $\hat{A}_f = A_f - G_f X_f = A_f - B_f K_f$ is identical to that of the left-corner block-matrix $S_{f_{11}}$, i.e.,

$$\hat{A}_f = \begin{bmatrix} -0.4270 & 648.4947 \\ -0.0103 & -3.4896 \end{bmatrix} \text{ and } S_{f_{11}} = \begin{bmatrix} -1.9583 & -3.0017 \\ 1.4431 & -1.9583 \end{bmatrix},$$

precisely

$$\Lambda(S_{f_{11}}) = \Lambda(\hat{A}_f) = \{-1.9583 \pm 2.0813\jmath\}.$$

Questions? vs. Answers!

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